Venkata Lakshmi Parimala Pasupuleti

Healthcare AI Engineer · Agentic Workflows · LLM Evaluation Harnesses · FHIR · HIPAA · LangGraph · Python · FastAPI · Production AI

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PROFESSIONAL SUMMARY

Senior software engineer with 8 years shipping production systems and hands-on agentic AI expertise built on real healthcare compliance and regulated-data pipelines. Built **Leasel** - a production **8-agent LangGraph StateGraph** for healthcare compliance document processing: structured extraction from unstructured PDFs, rule-based eligibility checking, citation-grounded RAG (hybrid TF-IDF + semantic reranking), conditional HITL escalation, and a **5-metric CI regression harness** running against 1,240 test cases. The domain analog to insurance authorization: extract → validate against regulatory rules → route or auto-decide. Built **AgentProbe** - an **evaluation harness** for non-deterministic LLM systems: golden datasets, LLM-as-judge scoring, online + offline eval loops, latency and hallucination monitoring - the observability infrastructure to keep LLM-centered production systems reliable as traffic grows. Built **AI on FHIR** - NLP-to-FHIR query translation with HIPAA-compliant RBAC across real medical records. At Entrata (8yr): owned regulated data pipelines for 7 years in a domain where a wrong output affects thousands of patients' eligibility - the same failure mode as healthcare AI. Python, FastAPI, LangGraph, LLM APIs (OpenAI/Anthropic/Groq/Ollama), Docker, Kubernetes, AWS. Case study: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Agentic Workflows | LLM APIs (OpenAI / Anthropic / Groq) | LangGraph / LangChain | HITL / Human-in-the-Loop | Evaluation Harnesses | LLM-as-Judge Scoring | CI Regression Testing | Model Deployment & Monitoring | Observability Infrastructure | Data Pipelines | RAG Pipelines / Hybrid Search | FHIR / HL7 | HIPAA / PHI Compliance | Healthcare AI | Full-Stack AI Systems | Python / FastAPI | React / TypeScript | Prototype to Production | System Design / Backend Engineering | AWS (S3 / Lambda / Bedrock) | Docker / Kubernetes | CI/CD (GitHub Actions) | PostgreSQL / Redis / SQLite | RabbitMQ / Celery | Multi-Tenant Architecture

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building ML models and data analysis pipelines; integrating AI components with backend/frontend teams; surfacing insights from raw business data into application-facing APIs and reporting layers.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → export) with risk-aware routing - compliant missions export to QGroundControl; non-compliant blocked with cited FAA regulation.
- Implemented a **RAG layer** over FAA Part 107 corpus with deterministic guardrails; built an MCP server enabling downstream tool integration via Model Context Protocol.

George Mason University

Aug 2024 - Aug 2025

Research Associate / Graduate Teaching Assistant — Virginia, USA

- Fine-tuned a custom classification model using HuggingFace Transformers + PyTorch; deployed inference API on Quart (async Python); bridged ML research to production inference - presented model design, training methodology, and evaluation results to faculty.
- Built multi-source prediction models (Python, R) on 190+ country datasets - R^2=0.977; applied Isolation Forest anomaly detection across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Owned regulated data pipelines for 7 years** - HUD income-limit ingestion (few thousand federal rows → ~180,000 domain records powering eligibility, rent, and HAP calculations for 100K+ housing units); built baseline validation, reconciliation, and rollback procedures for every annual cycle; zero compliance violations.
- **Built RabbitMQ-based async processing** for gross-rent workflow automation - designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual schedule changes.
- **Led AWS S3 document migration with 100% PHI compliance** - Libsodium-encrypted pipelines; zero security incidents across full resident data migration; HIPAA-compliant multi-tenant data isolation for 100K+ MAU.
- **Optimized 200+ SQL queries (30% performance gain)**; shipped 20+ full-stack production modules across 120+ Agile sprints; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual workload via Python/Bash automation.
- Mentored 5+ engineers; conducted 300+ peer code reviews; built internal debugging tools cutting debug time by 40%.

Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built property management and compliance modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration of client documents; income limits import (1.6M HUD/Tax Credit records across 5 years).

PROJECTS

AI on FHIR: NLP-Driven Medical Records Query System

FHIR · HIPAA · Clinical NLP · Healthcare AI

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural-language queries (EN/ES) into **FHIR API calls** using LLMs, BERT, and embeddings. Enforced **HIPAA-compliant RBAC** across structured medical records - PHI access controlled at the data layer. Multilingual clinical NLP using spaCy + BERT; containerized with Docker; supports structured clinical workflow queries including revenue cycle records.

Flask · React · FHIR APIs · HIPAA RBAC · LLMs · BERT · spaCy · Embeddings · Docker · 2024

LeaseI: Agentic AI Support Investigation Platform

LangGraph · 8-Agent · HITL · Eval Harness · Healthcare Compliance Analog

Production agentic system for healthcare compliance document processing - same problem structure as insurance authorization: extract structured data from unstructured PDFs → check against regulatory rules → route to human or auto-decide. **8-agent LangGraph StateGraph** with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation); **conditional HITL gate** when confidence is too low; hybrid RAG (TF-IDF + semantic reranking) with citation grounding; LLM extracts and reasons, deterministic logic makes every regulated call. Eval harness: 93.6% precision, 87.9% F1 on 1,240 test cases; pass@1 81.7% via CI regression gate.

Python · LangGraph (8-agent) · FastAPI · TF-IDF · Semantic Reranking · OCR · SQLite · 2026

AgentProbe: Open-Source AI Evaluation Toolkit

Eval Harness · LLM-as-Judge · CI Regression · Observability

Evaluation harness for non-deterministic LLM-centered production systems - 15-endpoint FastAPI backend, **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated golden-dataset regression harness, LLM-as-judge scoring, model routing across OpenAI/Anthropic/Groq/Ollama, and **online + offline evaluation loops**. Designed as the observability infrastructure that keeps LLM systems reliable as traffic grows. Used to regression-test LeaseI across 1,240+ compliance test cases. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · OpenAI / Anthropic / Groq / Ollama · 2025

EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017